

D 92460–A

(Pages : 6)

Name.....

Reg. No.....

**FIFTH SEMESTER U.G. DEGREE (SPECIAL) EXAMINATION
NOVEMBER 2020**

(CUCBCSS—UG)

Mathematics

MAT 5B 08—DIFFERENTIAL EQUATIONS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 5B 08—DIFFERENTIAL EQUATIONS

(Multiple Choice Questions for SDE Candidates)

1. Which of the following is a linear differential equation ?

(A) $y'' + (y')^2 = \sin x.$

(B) $(y')^2 + 3y = e^x.$

(C) $y'' + 3y' + y = 0.$

(D) $(y')^2 + (y')^3 + e^x = 0.$

2. Which of the following is a separable differential equation ?

(A) $\frac{dy}{dx} = \frac{x^2}{1 - y^2}.$

(B) $\frac{dy}{dx} = \frac{x + y}{x - y}.$

(C) $\frac{dy}{dx} + (\sin x)y = e^x.$

(D) $\left(\frac{dy}{dx}\right)^2 + (\sin x)y = 0.$

3. The general solution of the differential equation $2x(3x + y - ye^{-x^2})dx + (x^2 + 3y^2 + e^{-x^2})dy = 0$ is :

(A) $x^2y + ye^{-x^2} + 2x^3 + y^3 = C.$

(B) $x^2y^2 + ye^{x^2} + 2x + y^2 = C.$

(C) $xy + ye^{-x^2} + y^2 = C.$

(D) $xy^2 + y + 2x^3e^{-x^2} + y^3 = C.$

4. Which of the following is an initial value problem ?

(A) $y' + y = 0, y(0) = y'(0) = 0.$

(B) $y' + y = 0, y(0) = y(1) = 0.$

(C) $y'' + y = 0, y(0) = 0, y(1) = 1..$

(D) $y''' + y = 0, y(0) = 0, y(2) = 4.$

5. Which of the following is a boundary value problem :

(A) $y' + y = 0, y(0) = 1, y'(0) = 0.$

(B) $y'' + 5y = 0, y(0) = 1, y'(0) = 3.$

(C) $x^2y'' + xy' + y = 0, y(0) = 0, y(1) = 2.$

(D) $y'' + y'' + y = 0, y(0) = y'(0) = y''(0) = 0.$

6. The general solution of the differential equation $y' = \cos x$ is :

(A) $y = \sin x.$

(B) $y = \cos x.$

(C) $y = C \sin x.$

(D) $y = \sin(x) + C.$

7. If $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$ is a function of x only, then an integrating factor of $Mdx + Ndy = 0$ is :

(A) $\mu(x) = \exp \left[\int \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dx \right].$

(B) $\mu(x) = \exp \left[\int \frac{1}{N} \left(\frac{\partial M}{\partial y} + \frac{\partial N}{\partial x} \right) dx \right].$

(C) $\mu(x) = \int \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dx.$

(D) $\mu(x) = \int \frac{1}{N} \left(\frac{\partial M}{\partial x} - \frac{\partial N}{\partial y} \right) dx.$

8. If $\frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)$ is a function of y only, then an integrating factor of the differential equation $Mdx + Ndy = 0$ is :

(A) $\mu(x) = \exp \left[\int \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy \right].$

(B) $\mu(x) = \exp \left[\int \frac{1}{M} \left(\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y} \right) dy \right].$

(C) $\mu(x) = \int \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy.$

(D) $\mu(x) = \int \frac{1}{M} \left(\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y} \right) dy.$

9. An integrating factor of the differential equation $\frac{dx}{dy} + P(x) = Q(x)$ is :

(A) $e^{\int p dx}.$

(B) $e^{-\int p dx}.$

(C) $e^{\int p^2 dx}.$

(D) $e^{\int (P+Q) dx}.$

10. A mathematical model of an object falling in the atmosphere near the surface of earth is given by :

(A) $m \frac{dv}{dt} = mg - rv.$

(B) $m \frac{d^2v}{dt^2} = mg - rv.$

(C) $\frac{dv}{dt} = mg.$

(D) None of these.

11. The differential equation $y'' - 5y' + 6y = 0$ has :

(A) Two linearly independent solutions.

(B) Three linearly independent solutions.

(C) Four linearly independent solutions.

(D) Infinite number of linearly independent solution.

12. The general solution of the differential equation $(D^2 - 4D + 4)y = 0$ is :

(A) $(c_0 + c_1x)e^{2x}.$

(B) $(c_0 - c_1x)e^{2x}.$

(C) $c_1e^x c_2e^{-2x}.$

(D) $c_1e^{2x} + c_2e^{2x}.$

13. The characteristic roots of the differential equation $(D^2 - 2D)y = 4x^2 + 2x + 3$ are :

(A) $\lambda = 0, \lambda = -2.$

(B) $\lambda = 1, \lambda = 3.$

(C) $\lambda = 0, \lambda = 2.$

(D) $\lambda = 1, \lambda = -2.$

14. The Laplace transform of the unit step function $u_{(t-a)}$ is :

(A) $e^{-as}.$

(B) $e^{-as}/s.$

(C) $e^{as}/s.$

(D) $e^{-as}/s^2.$

15. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{f(at)\} =$

(A) $\frac{1}{a} F(s/a).$

(B) $F(s/a).$

(C) $F(a/s).$

(D) $F(s).$

16. $\int_0^{\infty} \frac{\sin t}{t} dt =$

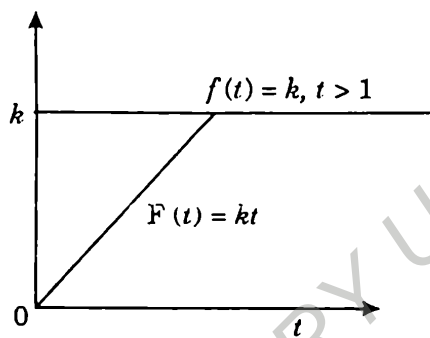
(A) $\frac{\pi}{4}.$

(B) $\frac{\pi}{8}.$

(C) $\frac{\pi}{2}.$

(D) None of these.

17. The Laplace transform of the function whose graph shown below is :



(A) $\frac{k}{s^2} (1 - e^{-s}).$

(B) $\frac{k}{s} (1 - e^{-s}).$

(C) $\frac{1}{s} (1 - e^{-s}).$

(D) None of these.

18. $\mathcal{L}\{\sin at\} =$

(A) $\frac{a}{s^2 - a^2}.$

(B) $\frac{a}{s^2 + a^2}.$

(C) $\frac{s}{s^2 - a^2}.$

(D) $\frac{s}{s^2 + a^2}.$

19. $\mathcal{L}\{t^n\} =$

(A) $\frac{n!}{s^n}$.

(B) $\frac{(n+1)!}{s^n}$.

(C) $\frac{n!}{s^n + 1}$.

(D) $\frac{1}{s^n}$.

20. The solution to the problem :

$$\alpha^2 u_{xx} = u_{tt}, 0 \leq x \leq L$$

$$\begin{cases} u(0, t) = 0 \\ u(L, t) = 0 \end{cases} \quad \begin{cases} u(x, 0) = f(x) \\ u_t(x, 0) = 0 \end{cases} \quad \text{is given by}$$

(A) $u(x, t) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{L}\right) \cos\left(\frac{n\pi t}{L}\right)$.

(B) $u(x, t) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi t}{L}\right) \cos\left(\frac{n\pi x}{L}\right)$.

(C) $u(x, t) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{L}\right)$.

(D) $u(x, t) = \sum_{n=1}^{\infty} b_n \cos\left(\frac{n\pi x}{L}\right)$.